

Project Concept Note: MedFlow AI

Executive Summary

MedFlow AI is an intelligent hospital triage and scheduling portal designed to streamline patient intake and clinical routing. By decoupling manual patient operations from automated AI-driven workflows, MedFlow AI ensures efficient, accurate, and safe patient management.

The Challenge

Healthcare environments frequently face critical inefficiencies:

- **Triage Bottlenecks:** Manual intake processes delay critical care.
- **Routing Inaccuracies:** Misclassification of emergency or psychiatric cases leads to poor outcomes.
- **Scheduling Fragmentation:** Decentralized, manual doctor scheduling prevents optimal resource allocation.
- **UI/UX Limitations:** Legacy software interfaces increase cognitive load for administrative staff.

Proposed Solution

MedFlow AI optimizes clinical operations through a two-layered architecture:

- **The Brain (Backend):** A LangGraph-powered state machine utilizing Llama 3.3 via Groq API. It enforces strict safety guardrails for crisis and emergency cases while using LLM reasoning for standard patient classification.
- **The Portal (Frontend):** A responsive, glassmorphic dashboard built on Vanilla HTML/CSS/JS, facilitating real-time scheduling, queue monitoring, and staff log management.

System Workflow

1. **Intake:** Patient data including symptoms and clinical history is securely collected.
2. **Safety Routing:** Crisis and emergency indicators trigger deterministic bypasses to priority wards, while general queries are processed via LLM reasoning.
3. **Smart Scheduling:** The system performs automated lookup in a doctor database, identifies the earliest availability, and logs the appointment.
4. **Visual Updates:** Results are dynamically displayed via a real-time clinical interface, providing clear classification reasoning and scheduling details.

Technology Stack

- **Agent Engine:** LangGraph, Langchain-core, Llama 3.3 70B (Groq).
- **Middleware:** FastAPI and Uvicorn.
- **Data Persistence:** Localized CSV data structures managed by Pandas.
- **Interface:** Vanilla HTML5, CSS3, and JavaScript with custom SVGs and responsive layouts.

Value Proposition

- **Enhanced Patient Safety:** Deterministic emergency/crisis detection ensures immediate triage prioritization.
- **Efficiency Gains:** Automated scheduling eliminates manual conflicts and reduces administrative wait times.
- **Centralized Coordination:** Integrated doctor directories and queue management simplify staff workflows.
- **Modern Experience:** A refined glassmorphic UI reduces visual fatigue and improves operational clarity.

Roadmap

- **EHR Integration:** Connecting with standardized Hospital Electronic Health Record (EHR) systems.
- **Proactive Notifications:** Automated patient updates via SMS and email.
- **Telehealth Expansion:** Future scope for integrating remote consultation tools directly into the dashboard.